AQUASENSE

Water Level Indicator

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Contents

INTRODUCTION

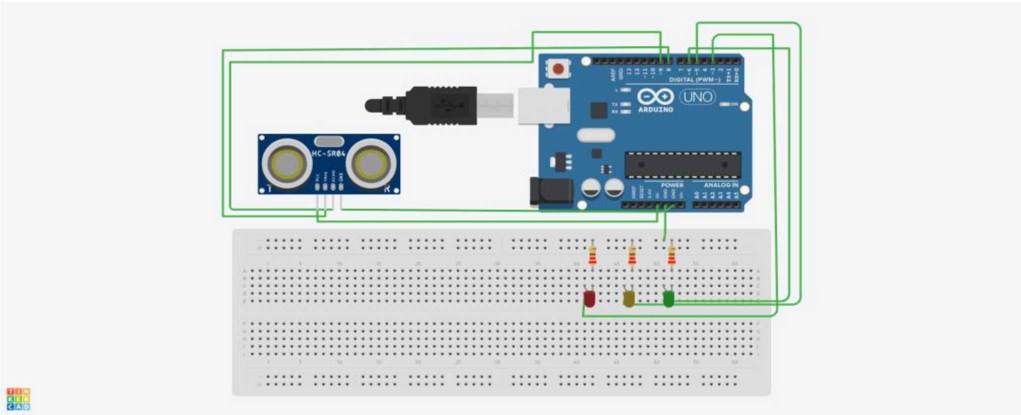
The **Smart Water Tank Level Monitoring System (Aqua Sense)** is designed and simulated using **Tinkercad Circuits** to monitor the water level inside a storage tank. The project aims to demonstrate the working of a smart water level monitoring system through virtual circuit simulation.

The system uses an **HC-SR04 ultrasonic sensor** to measure the distance between the sensor and the water surface. The measured distance is processed by an **Arduino Uno**, which determines the water level condition inside the tank. Based on the detected water level, the tank status is indicated using **LED indicators**, providing a simple and clear visual representation of the water level.

Components Required

Component	Quantity	Purpose
Arduino Uno	1	Main controller
HC-SR04 Ultrasonic Sensor	1	Measures water level distance
LEDs	3	Indicates tank status
Resistors	3	LED protection
Breadboard	1	Circuit connection
Connecting wires		

Circuit Diagram



Working

The HC-SR04 ultrasonic sensor continuously measures the distance between the sensor and the water surface. The Arduino processes the sensor readings and determines the water level condition. Based on the calculated level, LEDs are turned ON to indicate the tank status.

Green LED → Tank has sufficient water

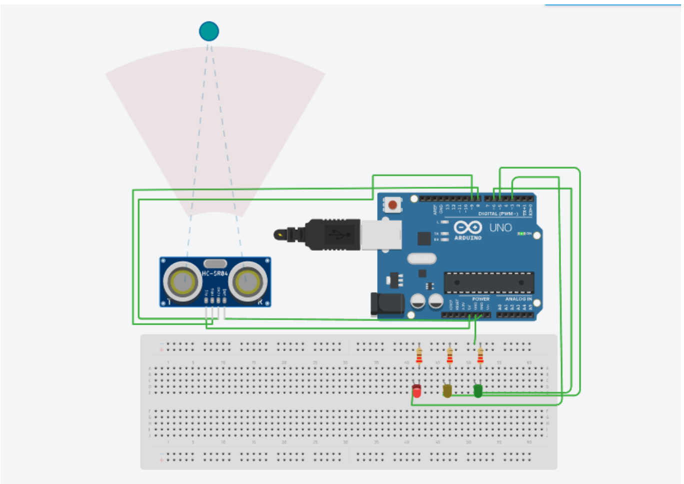
Yellow LED → Medium water level

Red LED → Low water level

Algorithm

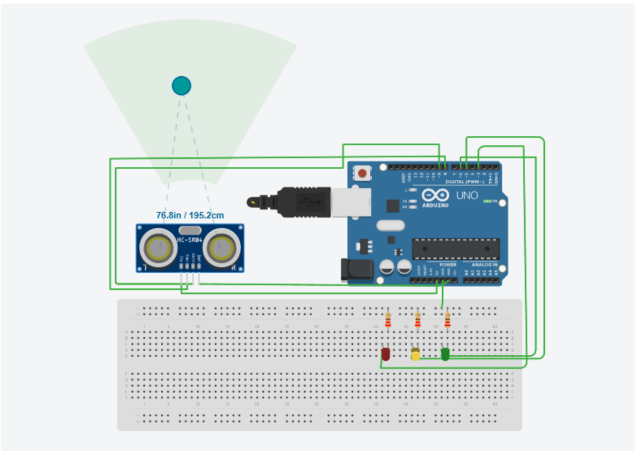
1. **Start the System**
   The Arduino initializes the ultrasonic sensor and LED pins required for the operation.
2. **Initialize Components**
   The HC-SR04 ultrasonic sensor is configured to send and receive ultrasonic signals, and the LEDs are set as output indicators.
3. **Measure Distance**
   The ultrasonic sensor sends a sound pulse towards the water surface and receives the reflected echo. The Arduino calculates the distance between the sensor and the water surface using the time taken for the echo to return.
4. **Determine Water Level Condition**
   The calculated distance is compared with predefined threshold values to identify the water level condition inside the tank.
5. **Indicate Tank Status**
   Based on the detected water level, the corresponding LED is turned ON to indicate the tank status:
   - High water level → Green LED ON
   - Medium water level → Yellow LED ON
   - Low water level → Red LED ON
6. **Repeat Monitoring Process**
   The system continuously repeats the measurement and indication process to provide real-time water level monitoring.
7. **End**
   The process continues as long as the system remains powered on.

SIMULATION RESULTS



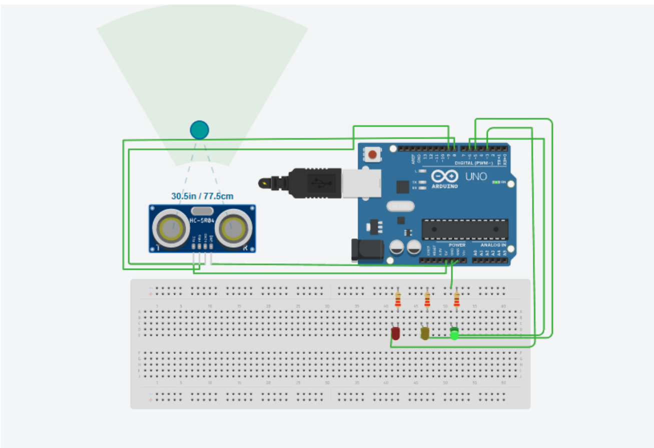
Case 1: Low Water Level Indication
(Red LED Blinking)

In the first simulation case, the water level inside the tank is detected as **low** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor data and identifies that the water level has reached below the predefined threshold value. As an indication of the low water level condition, the **red LED starts blinking**



Case 2: Medium Water Level Indication (Yellow LED Blinking)

In the second simulation case, the water level inside the tank is detected as **medium** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor readings and determines that the water level is within the medium range based on the predefined threshold values.



Case 3: High Water Level Indication (Green LED Blinking)

In the third simulation case, the water level inside the tank is detected as **high** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor readings and identifies that the water level has reached the high level range according to the predefined threshold values.

As an indication of the high-water level condition, the **green LED starts blinking**.

CONCLUSION

The **Aqua Sense – Smart Water Tank Level Monitoring System** was successfully designed and simulated using **Tinkercad Circuits**. The system uses an **HC-SR04 ultrasonic sensor** to measure the distance between the sensor and the water surface and determines the water level inside the tank using an **Arduino Uno**.

The tank status is effectively indicated using **LED indicators**, where different LEDs represent different water level conditions.